Que 1 →
● Create 1 Public Docker Hub registry named flask_webapp_yourname.
● Clone below repository on your system.
https://github.com/mmumshad/simple-webapp-docker.git
● Initialize a local repository & copy the code from above repo to your local repository in
your working branch.
● Once code is copied to the local repository, build the docker image & add meaningful
tags with version 1 and push to Docker Hub registry.
● Once Images are pushed to Docker hub registries, create a directory named kube.
Inside the kube directory create deployement.yaml file with 3 replication , labels app:
flask-webapp , containerPort: 8080 and add the image that you have pushed in Docker
Hub registry.
● Create one service.yaml file with type nodeport & select flask-webapp with port 8080 &
targetPort 8080 with any nodePort between range 30000-32768.
● Once a service is created try accessing the web page in the browser as below. (30011 is
nodeport mentioned in service.yaml). Meanwhile open app.py from your code to understand paths & output.
http://master_ip:30011/
http://master_ip:30011/how are you
● Now , update the app.py from your code and add below route above if __name__ ==
"__main__" line
@app.route('/Who are you')
def cloudethix():
return 'Yes, I am cloudethix, and You !!!'
● Once the file is updated , rebuild the docker image & add meaningful tags with version 2
and push to Docker Hub registry.
● Now we have the latest docker image in repo, It's time to roll out a new image. Roll out
the new Image with all three ways one by one.
1. With kubectl set command
2. With kubectl edit deployment
3. With deployment.yaml file modification.
● Run the # kubectl rollout command to check status and history.
● Note:- Once above step 1 is done , run # kubectl rollout undo deployment command to
rollback the change and then try a second way of rollout.
● In the browser run all three routes & notice the changes.
http://master_ip:30011/
http://master_ip:30011/how are you
http://master_ip:30011/Who are you
● Once done with all above steps , commit all the changes to the remote repository.
● Capture the snap and prepare a well formatted document.
---
achal@LAPTOP-# git clone https://github.com/mmumshad/simple-webapp-docker.git
Cloning into 'simple-webapp-docker'...
remote: Enumerating objects: 14, done.
remote: Counting objects: 100% (7/7), done.

```
remote: Compressing objects: 100% (5/5), done.
remote: Total 14 (delta 3), reused 2 (delta 2), pack-reused 7 (from 1)
Receiving objects: 100% (14/14), done.
Resolving deltas: 100% (3/3), done.
-----------------------------------------------------------------
achal@LAPTOP-# cp -r simple-webapp-docker/ local-repo/
achal@LAPTOP-HPUOTJF1:Assignment-practice-3$ cd local-repo/
achal@LAPTOP-HPUOTJF1:local-repo$ ls
simple-webapp-docker
-----------------------------------------------------------------
achal@LAPTOP-# podman build -t docker.io/achalpadol/flask_webapp_achal:v1 .
STEP 1/5: FROM ubuntu:20.04
Resolved "ubuntu" as an alias
(/etc/containers/registries.conf.d/shortnames.conf)
Trying to pull docker.io/library/ubuntu:20.04...
Getting image source signatures
Copying blob 13b7e930469f done   |
Copying config b7bab04fd9 done   |
Writing manifest to image destination
STEP 2/5: RUN apt-get update && apt-get install -y python3 python3-pip
Get:1 http://security.ubuntu.com/ubuntu focal-security InRelease [128 kB]
Get:2 http://archive.ubuntu.com/ubuntu focal InRelease [265 kB]
Get:3 http://security.ubuntu.com/ubuntu focal-security/universe amd64 Packages
[1300 kB]
Get:4 http://security.ubuntu.com/ubuntu focal-security/restricted amd64
Packages [4710 kB]
Get:5 http://security.ubuntu.com/ubuntu focal-security/main amd64 Packages
[4368 kB]
Get:6 http://archive.ubuntu.com/ubuntu focal-updates InRelease [128 kB]
Get:7 http://archive.ubuntu.com/ubuntu focal-backports InRelease [128 kB]
Get:8 http://security.ubuntu.com/ubuntu focal-security/multiverse amd64
Packages [32.5 kB]
Get:9 http://archive.ubuntu.com/ubuntu focal/main amd64 Packages [1275 kB]
-----------------------------------------------------------------
achal@LAPTOP-# podman images
REPOSITORY                                          TAG               IMAGE
ID      CREATED         SIZE
docker.io/achalpadol/flask_webapp_achal             v1
949f9230f443  23 seconds ago  480 MB
-----------------------------------------------------------------
achal@LAPTOP-# docker push docker.io/achalpadol/flask_webapp_achal:v1
Getting image source signatures
Copying blob 5a8ac36787a8 done   |
Copying blob 8168e02bd347 done   |
Copying blob 721125e8a881 done   |
Copying blob 13b7e930469f skipped: already exists
Copying config 949f9230f4 done   |
Writing manifest to image destination
-----------------------------------------------------------------
achal@LAPTOP-# cat deployment.yml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: flask-webapp-1
  labels:
    app: flask-webapp-1
spec:
  replicas: 3
  selector:
    matchLabels:
```

```
      app: flask-webapp-1
  template:
    metadata:
      labels:
        app: flask-webapp-1
    spec:
      containers:
        - name: flask-webapp-1
          image: docker.io/achalpadol/flask_webapp_achal:v1
          ports:
            - containerPort: 8080
-----------------------------------------------------------------------
achal@LAPTOP-#:kube$ kubectl get deployment
NAME             READY   UP-TO-DATE   AVAILABLE   AGE
flask-webapp-1   3/3     3            3           2m29s
-----------------------------------------------------------------------
achal@LAPTOP-#:kube$ kubectl get pod
NAME                              READY   STATUS              RESTARTS   AGE
flask-webapp-1-74458c66d7-6dt2p   1/1     Running             0          2m42s
flask-webapp-1-74458c66d7-hk4nd   1/1     Running             0          2m42s
flask-webapp-1-74458c66d7-knwd8   1/1     Running             0          2m42s
-----------------------------------------------------------------------
achal@LAPTOP-# cat service.yml
apiVersion: v1
kind: Service
metadata:
  name: flask-webapp-service
spec:
  type: NodePort
  selector:
    app: flask-webapp-1
  ports:
    - port: 8080
      targetPort: 8080
      nodePort: 30011
-----------------------------------------------------------------------
achal@LAPTOP-#:kube$ kubectl get svc
NAME                   TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)
AGE
flask-webapp-service   NodePort    10.100.2.192   <none>        8080:30011/TCP
4m30s
-----------------------------------------------------------------------
achal@LAPTOP-#:kube$ curl http://192.168.49.2:30011
Welcome!

achal@LAPTOP-#:kube$ curl "http://192.168.49.2:30011/how%20are%20you"
I am good, how about you?
-----------------------------------------------------------------------
achal@LAPTOP-# cat app.py
import os
from flask import Flask
app = Flask(__name__)

@app.route("/")
def main():
    return "Welcome!"

@app.route('/how are you')
def hello():
    return 'I am good, how about you?'
```

```
if __name__ == "__main__":
    app.run()
-------------------------------------------------------------------
# updated app.py

achal@LAPTOP-# cat app.py
import os
from flask import Flask
app = Flask(__name__)

@app.route("/")
def main():
    return "Welcome!"

@app.route('/how are you')
def hello():
    return 'I am good, how about you?'

@app.route('/Who are you')
def cloudethix():
    return 'Yes, I am cloudethix, and You !!!'

if __name__ == "__main__":
    app.run()
-------------------------------------------------------------------
achal@LAPTOP-# podman build -t docker.io/achalpadol/flask_webapp_achal:v2 .

STEP 1/5: FROM ubuntu:20.04
STEP 2/5: RUN apt-get update && apt-get install -y python3 python3-pip
--> Using cache
eb374c6995cc5c9da545c53818fb15b63b9d3ea3cda6309be11299a130876107
--> eb374c6995cc
STEP 3/5: RUN pip install flask
--> Using cache
f8d2a7f56585a18b5799133c76d7a2f3ed3afe3d4f84a013e73997307bfafd19
--> f8d2a7f56585
STEP 4/5: COPY app.py /opt/
--> ab831a5f60a7
STEP 5/5: ENTRYPOINT FLASK_APP=/opt/app.py flask run --host=0.0.0.0 --
port=8080
COMMIT docker.io/achalpadol/flask_webapp_achal:v2
--> 43ecf906970a
Successfully tagged docker.io/achalpadol/flask_webapp_achal:v2
43ecf906970a397be9d2c4c65836b34b26b8dde28c170c6c15059ee1e5068ebc
-------------------------------------------------------------------
achal@LAPTOP-# podman images

REPOSITORY                                          TAG               IMAGE
ID      CREATED        SIZE
docker.io/achalpadol/flask_webapp_achal             v2
43ecf906970a  4 seconds ago  480 MB
-------------------------------------------------------------------
achal@LAPTOP-# podman images | grep flask_webapp
docker.io/achalpadol/flask_webapp_achal             v2
43ecf906970a  About a minute ago  480 MB
docker.io/achalpadol/flask_webapp_achal             v1
949f9230f443  14 hours ago        480 MB
-------------------------------------------------------------------
achal@LAPTOP-# podman push docker.io/achalpadol/flask_webapp_achal:v2
```

```
Getting image source signatures
Copying blob 6ac90cd9bb76 done   |
Copying blob 13b7e930469f skipped: already exists
Copying blob d59147826128 skipped: already exists
Copying blob bde202e56e37 skipped: already exists
Copying config 43ecf90697 done   |
Writing manifest to image destination
-----------------------------------------------------------------------
achal@LAPTOP-# kubectl get deployment
NAME             READY   UP-TO-DATE   AVAILABLE   AGE
flask-webapp-1   3/3     3            3           135m
-----------------------------------------------------------------------
achal@LAPTOP-# kubectl describe deployment flask-webapp-1 | grep Image
    Image:         docker.io/achalpadol/flask_webapp_achal:v1
-----------------------------------------------------------------------
achal@LAPTOP-# kubectl set image deployment/flask-webapp-1 flask-webapp-
1=docker.io/achalpadol/flask_webapp_achal:v2
deployment.apps/flask-webapp-1 image updated
-----------------------------------------------------------------------
achal@LAPTOP-# kubectl describe deployment flask-webapp-1 | grep Image
    Image:         docker.io/achalpadol/flask_webapp_achal:v2
-----------------------------------------------------------------------
achal@LAPTOP-# kubectl rollout status deployment/flask-webapp-1
deployment "flask-webapp-1" successfully rolled out
-----------------------------------------------------------------------
achal@LAPTOP-# kubectl rollout history deployment/flask-webapp-1
deployment.apps/flask-webapp-1
REVISION  CHANGE-CAUSE
1         <none>
2         <none>
-----------------------------------------------------------------------
achal@LAPTOP-# kubectl rollout undo deployment/flask-webapp-1
Warning: resource deployments/flask-webapp-1 was previously managed with
'kubectl apply'. Rolling back will not update the kubectl.kubernetes.io/last-
applied-configuration annotation, which may cause unexpected behavior on
future 'kubectl apply' operations. Consider using 'kubectl apply' with your
previous configuration file instead.
deployment.apps/flask-webapp-1 rolled back
-----------------------------------------------------------------------
achal@LAPTOP-# kubectl describe deployment flask-webapp-1 | grep Image
    Image:         docker.io/achalpadol/flask_webapp_achal:v1
-----------------------------------------------------------------------
# Edit deployment

achal@LAPTOP-# kubectl edit deployment flask-webapp-1
deployment.apps/flask-webapp-1 edited

achal@LAPTOP-# kubectl rollout status deployment/flask-webapp-1
deployment "flask-webapp-1" successfully rolled out
achal@LAPTOP-# kubectl rollout history deployment/flask-webapp-1
deployment.apps/flask-webapp-1
REVISION  CHANGE-CAUSE
3         <none>
4         <none>
-----------------------------------------------------------------------
# undo again to check third step

achal@LAPTOP-# kubectl rollout undo deployment/flask-webapp-1
Warning: resource deployments/flask-webapp-1 was previously managed with
'kubectl apply'. Rolling back will not update the kubectl.kubernetes.io/last-
```

```
applied-configuration annotation, which may cause unexpected behavior on
future 'kubectl apply' operations. Consider using 'kubectl apply' with your
previous configuration file instead.
deployment.apps/flask-webapp-1 rolled back

achal@LAPTOP-# kubectl describe deployment flask-webapp-1 | grep Image
    Image:         docker.io/achalpadol/flask_webapp_achal:v1
-----------------------------------------------------------------------
achal@LAPTOP-# kubectl rollout status deployment/flask-webapp-1
deployment "flask-webapp-1" successfully rolled out
-----------------------------------------------------------------------
achal@LAPTOP-# kubectl rollout history deployment/flask-webapp-1
deployment.apps/flask-webapp-1
REVISION  CHANGE-CAUSE
4         <none>
5         <none>
-----------------------------------------------------------------------
#with deployment.yaml file modification.

achal@LAPTOP-# cat deployment.yml
achal@LAPTOP-# cat deployment.yml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: flask-webapp-1
  labels:
    app: flask-webapp-1
spec:
  replicas: 3
  selector:
    matchLabels:
      app: flask-webapp-1
  template:
    metadata:
      labels:
        app: flask-webapp-1
    spec:
      containers:
        - name: flask-webapp-1
          image: docker.io/achalpadol/flask_webapp_achal:v2
          ports:
            - containerPort: 8080
-----------------------------------------------------------------------
achal@LAPTOP-# kubectl apply -f deployment.yml
deployment.apps/flask-webapp-1 configured
-----------------------------------------------------------------------
achal@LAPTOP-# kubectl rollout status deployment/flask-webapp-1
deployment "flask-webapp-1" successfully rolled out
-----------------------------------------------------------------------
achal@LAPTOP-# kubectl rollout history deployment/flask-webapp-1
deployment.apps/flask-webapp-1
REVISION  CHANGE-CAUSE
5         <none>
6         <none>
-----------------------------------------------------------------------
achal@LAPTOP-# curl http://192.168.49.2:30011
Welcome!

achal@LAPTOP-# curl "http://192.168.49.2:30011/how%20are%20you"
I am good, how about you?
```

```
achal@LAPTOP-# curl http://192.168.49.2:30011/Who%20are%20you
Yes, I am cloudethix, and You !!!
-----------------------------------------------------------------------
achal@LAPTOP-# git branch -M main
achal@LAPTOP-HPUOTJF1:local-repo$ git remote add origin
https://github.com/achalpadol/cloudethix-rollout-repo-2.git

achal@LAPTOP-# git remote -v
origin  https://github.com/achalpadol/cloudethix-rollout-repo-2.git (fetch)
origin  https://github.com/achalpadol/cloudethix-rollout-repo-2.git (push)

achal@LAPTOP-# git push -u origin main
Username for 'https://github.com': achalpadol
Password for 'https://achalpadol@github.com':
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 12 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 670 bytes | 670.00 KiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/achalpadol/cloudethix-rollout-repo-2.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
-----------------------------------------------------------------------
Que 2 →
● Download and install Lens & access your k8s cluster from Lens.
● Create nginx Pod and Nodeport service. Check the Pod logs from Lens.
● Check the service from lens. Also login to the pod shell using the lens.
● Take snaps and delete the resources that you have just created.
-----------------------------------------------------------------------
achal@LAPTOP-# minikube dashboard
🔌  Enabling dashboard ...
    ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
    ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡  Some dashboard features require the metrics-server addon. To enable all
features please run:

        minikube addons enable metrics-server

🤔  Verifying dashboard health ...
🚀  Launching proxy ...
🤔  Verifying proxy health ...
🎉  Opening http://127.0.0.1:33983/api/v1/namespaces/kubernetes-
dashboard/services/http:kubernetes-dashboard:/proxy/ in your default
browser...
👉  http://127.0.0.1:33983/api/v1/namespaces/kubernetes-
dashboard/services/http:kubernetes-dashboard:/proxy/
-----------------------------------------------------------------------
achal@LAPTOP-# cat pod.yml
apiVErsion: v1
kind: Pod
metadata:
  name: pod-nginx
  labels:
    app: app-nginx
spec:
  containers:
    - name: app-nginx
      image: nginx:latest
```

```
      ports:
        - containerPort: 80
-----------------------------------------------------------------------
achal@LAPTOP-# cat service.yml
apiVErsion: v1
kind: Service
metadata:
  name: NodePort-Service
spec:
  selector:
    app: app-nginx
  type: NodePort
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30007
-----------------------------------------------------------------------
Que 3 →
● Create 1 Public Docker Hub registry named cloudethix_configmap_yourname.
● Clone below repository on your system.
https://github.com/zembutsu/docker-sample-nginx.git
● Initialize a local repository & copy the code from above repo to your local
repository in
the working branch.
● Once code is copied , build a docker image from docker file and add
meaningful tags
and push to docker hub repository.
● Once Images are pushed to Docker hub registries, create a directory named
kube.
Inside the kube directory create deployement.yaml file with 3 replication ,
labels app:
frontend-webapp , containerPort: 80 and add the image that you have pushed in
Docker
Hub registry.
● Create one service.yaml file with type nodeport & select frontend-webapp pod
with port
80 & targetPort 80 with any nodePort between range 30000-32768.
● Once the service is created try accessing the web page in the browser as
below. Notice
the changes & take the snap.
● Now create a configmap.yaml file with below data & delete the deployment
that you have
created.
<html>
<body>
<h1> I am Cloudethix Team, Are you ?!! </h1>
Version: 1.1
</body>
</html>
● Then update the same deployment.yaml file and mount configmap as volume on
container using volumeMounts with mountPath /usr/share/nginx/html/
● Now it's time to create configmap & deployment. Once created , try to access
the
webpage in the browser & confirm that the index page is the same as we have in
configmap.
-----------------------------------------------------------------------
achal@LAPTOP-# git clone https://github.com/zembutsu/docker-sample-nginx.git
Cloning into 'docker-sample-nginx'...
remote: Enumerating objects: 22, done.
remote: Counting objects: 100% (12/12), done.
```

```
remote: Compressing objects: 100% (6/6), done.
remote: Total 22 (delta 7), reused 6 (delta 6), pack-reused 10 (from 1)
Receiving objects: 100% (22/22), done.
Resolving deltas: 100% (7/7), done.
-----------------------------------------------------------------
achal@LAPTOP-# podman build -t
docker.io/achalpadol/cloudethix_configmap_achal:v1 .

WARN[0000] "/" is not a shared mount, this could cause issues or missing
mounts with rootless containers
STEP 1/3: FROM nginx:alpine
STEP 2/3: COPY default.conf /etc/nginx/conf.d/
--> Using cache
090b497e1161956bc69c29ed05f63003fabe8615f06a8b8a09cc43f4e2783826
--> 090b497e1161
STEP 3/3: COPY index.html /usr/share/nginx/html/
--> Using cache
97ebac17772705f64c7fb37106bd497011c968ff7fdf48c4a0f6d8233a32ea5b
COMMIT docker.io/achalpadol/cloudethix_configmap_achal:v1
--> 97ebac177727
Successfully tagged docker.io/achalpadol/cloudethix_configmap_achal:v1
Successfully tagged docker.io/achalpadol/cloudethix-nginix-achal:v1
Successfully tagged localhost/achalpadol/cloudethix-nginix-achal:v1
Successfully tagged localhost/cloudethix-nginix-achal:v1
97ebac17772705f64c7fb37106bd497011c968ff7fdf48c4a0f6d8233a32ea5b
-----------------------------------------------------------------
achal@LAPTOP-# podman images | grep cloudethix_configmap
docker.io/achalpadol/cloudethix_configmap_achal      v1
97ebac177727  24 hours ago   64.2 MB
-----------------------------------------------------------------
achal@LAPTOP-# podman push docker.io/achalpadol/cloudethix_configmap_achal:v1
Getting image source signatures
Copying blob 0d62d88506ba skipped: already exists
Copying blob d94291c26261 skipped: already exists
Copying blob 1ed8e39a7434 skipped: already exists
Copying blob 8cdfe4f23778 skipped: already exists
Copying blob 0ea935727878 skipped: already exists
Copying blob 55afa1ecc21d skipped: already exists
Copying blob fb597529c916 skipped: already exists
Copying blob da4dea1b00af skipped: already exists
Copying blob 0cc7f3a48f3f skipped: already exists
Copying blob 4b794759b385 skipped: already exists
Copying config 97ebac1777 done   |
Writing manifest to image destination
-----------------------------------------------------------------
achal@LAPTOP-# cat deployment.yml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: frontend-webapp-1
  namespace: cloudethix-achal
  labels:
    app: frontend-webapp-1
spec:
  replicas: 3
  selector:
    matchLabels:
      app: frontend-webapp-1
  template:
    metadata:
```

```
      labels:
        app: frontend-webapp-1
    spec:
      containers:
        - name: frontend-webapp-1
          image: docker.io/achalpadol/cloudethix_configmap_achal:v1
          ports:
            - containerPort: 80
-----------------------------------------------------------------------
achal@LAPTOP-# cat service.yml
apiVersion: v1
kind: Service
metadata:
  name: frontend-webapp-service
  namespace: cloudethix-achal
spec:
  type: NodePort
  selector:
    app: frontend-webapp-1
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30012
-----------------------------------------------------------------------
achal@LAPTOP-# minikube service frontend-webapp-service --url -n cloudethix-
achal
http://127.0.0.1:40131
❗  Because you are using a Docker driver on linux, the terminal needs to be
open to run it.
-----------------------------------------------------------------------
^Cachal@LAPTOP-# kubectl get pod
NAME                                  READY   STATUS             RESTARTS
AGE
frontend-webapp-1-74bc7cdf58-7bdbm   1/1     Running            0
5m25s
frontend-webapp-1-74bc7cdf58-rnrvq   1/1     Running            0
5m25s
frontend-webapp-1-74bc7cdf58-vxxmr   1/1     Running            0
5m25s
-----------------------------------------------------------------------
achal@LAPTOP-# kubectl get svc
NAME                      TYPE       CLUSTER-IP       EXTERNAL-IP   PORT(S)
AGE
frontend-webapp-service   NodePort   10.110.136.109   <none>
80:30012/TCP   5m5s
-----------------------------------------------------------------------
achal@LAPTOP-# curl http://127.0.0.1:40131
<!DOCTYPE html>
<html>
<head>
    <title>CloudEthiX Nginx</title>
</head>
<body>
    <h1>CloudEthiX</h1>
    <h2>Hostname: nginx</h2>
</body>
</html>
-----------------------------------------------------------------------
achal@LAPTOP-# cat configmap.yml
apiVersion: v1
```

```
kind: ConfigMap
metadata:
  name: frontend-html
  namespace: cloudethix-achal
data:
  index.html: |
    <html>
    <body>
    <h1> I am Cloudethix Team, Are you ?!! </h1>
    Version: 1.1
    </body>
    </html>
-----------------------------------------------------------------------
achal@LAPTOP-HPUOTJF1:kube$ kubectl apply -f configmap.yml
configmap/frontend-html created

^Cachal@LAPTOP-HPUOTJF1:kube$ kubectl get cm
NAME               DATA   AGE
frontend-html      1      3m41s
-----------------------------------------------------------------------
# updated deployment.yml
achal@LAPTOP-# cat deployment.yml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: frontend-webapp-1
  namespace: cloudethix-achal
  labels:
    app: frontend-webapp-1

spec:
  replicas: 3

  selector:
    matchLabels:
      app: frontend-webapp-1

  template:
    metadata:
      labels:
        app: frontend-webapp-1

    spec:
      containers:
        - name: frontend-webapp-1
          image: docker.io/achalpadol/cloudethix_configmap_achal:v1
          ports:
            - containerPort: 80

          volumeMounts:
            - name: nginx-html
              mountPath: /usr/share/nginx/html/

      volumes:
        - name: nginx-html
          configMap:
            name: frontend-html
-----------------------------------------------------------------------
achal@LAPTOP-# kubectl get deployment
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
```

```
frontend-webapp-1   3/3     3            3           7m54s
-----------------------------------------------------------------------
achal@LAPTOP-# minikube service frontend-webapp-service  --url -n cloudethix-
achal
http://127.0.0.1:33741
❗  Because you are using a Docker driver on linux, the terminal needs to be
open to run it.

achal@LAPTOP-HPUOTJF1:~$ curl http://127.0.0.1:33741
<html>
<body>
<h1> I am Cloudethix Team, Are you ?!! </h1>
Version: 1.1
</body>
</html>
-----------------------------------------------------------------------
Que 4 →
● Create 1 Public Docker Hub registry named
cloudethix_multicontainer_yourname.
● Clone below repository on your system.
https://github.com/janakiramm/Kubernetes-multi-container-pod.git
● Initialize a local repository & copy the code from above repo to your local
repository in
any of your working branches.
● Once code is copied , go to the Build directory and build docker image from
docker file
and add meaningful tags and push to docker hub repository.
● Now go to the deploy directory and notice the files.
● Here, web-pod-1.yml file will create the pod with two containers (Multi
container). Take a
note of lables , name of containers and ports. Also, please make sure you will
update the
python container image that you have pushed to your docker registry.
● Now, open web-svc.yml file and notice service Type , selectors & targetPort.
Apply the
file.
● Now open db-pod.yml & notice the lables , name , Image, containerPort and
apply the
file.
● Now open the db-svc.yml file and notice service Type , selectors &
targetPort. Apply the
file.
● Once above files are applied , Check that the Pods and Services are created
using
command line or lens.
● Now , from the command line run below urls & notice the changes.
# curl http://$NODE_IP:$NODE_PORT/init
Initialize the database with sample schema
● Now it's time to Insert some sample data. Make sure you will use correct
$NODE_IP:$NODE_PORT
# curl -i -H "Content-Type: application/json" -X POST -d '{"uid": "1",
"user":"John Doe"}' http://$NODE_IP:$NODE_PORT/users/add
# curl -i -H "Content-Type: application/json" -X POST -d '{"uid": "2",
"user":"Jane Doe"}' http://$NODE_IP:$NODE_PORT/users/add
# curl -i -H "Content-Type: application/json" -X POST -d '{"uid": "3",
"user":"Bill Colls"}' http://$NODE_IP:$NODE_PORT/users/add
# curl -i -H "Content-Type: application/json" -X POST -d '{"uid": "4",
"user":"Mike Taylor"}' http://$NODE_IP:$NODE_PORT/users/add
● Now access the data that we have added to database using below command.
# curl http://$NODE_IP:$NODE_PORT/users/1
```

```
● The second time you access the data, it appends '(c)' indicating that it is
pulled from the
Redis cache.
● Also, try to access mysql shell i.e db pod & run select * from the users
table. check
app.py for DB related information.
● Prepare proper documentation in brief & write start to end flow. Refer below
link if you
face any issues.
https://github.com/janakiramm/Kubernetes-multi-container-pod
--------------------------------------------------------------------
achal@LAPTOP-# git clone https://github.com/janakiramm/Kubernetes-multi-
container-pod.git
Cloning into 'Kubernetes-multi-container-pod'...
remote: Enumerating objects: 51, done.
remote: Total 51 (delta 0), reused 0 (delta 0), pack-reused 51 (from 1)
Receiving objects: 100% (51/51), 88.14 KiB | 247.00 KiB/s, done.
Resolving deltas: 100% (21/21), done.
achal@LAPTOP-HPUOTJF1:task-4$ ls
Kubernetes-multi-container-pod
--------------------------------------------------------------------
achal@LAPTOP-# podman build -t
docker.io/achalpadol/cloudethix_multicontainer_achal:v1 .
STEP 1/3: FROM python:2.7-onbuild
Resolved "python" as an alias
(/etc/containers/registries.conf.d/shortnames.conf)
Trying to pull docker.io/library/python:2.7-onbuild...
Getting image source signatures
Copying blob 0c3de61682aa done   |
Copying blob 46dde23c37b3 done   |
Copying blob d660b1f15b9b done   |
Copying blob e7428f935583 done   |
Copying blob 56f10ddf1173 done   |
Copying blob 6ebaeb074589 done   |
Copying blob 4473537c621d done   |
Copying blob 3106f7df3d1c done   |
Copying blob 3de1c6ceef68 done   |
Copying config 3f246dd60a done   |
Writing manifest to image destination
STEP 2/6: COPY requirements.txt /usr/src/app/
--> 88f14b23e568
STEP 3/6: RUN pip install --no-cache-dir -r requirements.txt
Collecting flask (from -r requirements.txt (line 1))
  Downloading https://files.pythonhosted.org/packa
--------------------------------------------------------------------
achal@LAPTOP-# podman push
docker.io/achalpadol/cloudethix_multicontainer_achal:v1
Getting image source signatures
Copying blob 6ebaeb074589 skipped: already exists
Copying blob 0c3de61682aa skipped: already exists
Copying blob 46dde23c37b3 skipped: already exists
Copying blob d660b1f15b9b skipped: already exists
Copying blob 56f10ddf1173 skipped: already exists
Copying blob e7428f935583 skipped: already exists
Copying blob 4f0e84c5ce6c done   |
Copying blob 4cf08394a795 done   |
Copying blob 0ba33796ade2 done   |
Copying blob 3de1c6ceef68 skipped: already exists
Copying blob 3106f7df3d1c skipped: already exists
Copying blob 4473537c621d skipped: already exists
```

```
Copying config 7997bb4680 done   |
Writing manifest to image destination
--------------------------------------------------------------------
achal@LAPTOP-# cat web-pod-1.yml
apiVersion: "v1"
kind: Pod
metadata:
  name: web1
  labels:
    name: web
    app: demo
spec:
  containers:
    - name: redis
      image: redis
      ports:
        - containerPort: 6379
          name: redis
          protocol: TCP
    - name: python
      image: docker.io/achalpadol/cloudethix_multicontainer_achal:v1
      env:
        - name: "REDIS_HOST"
          value: "localhost"
      ports:
        - containerPort: 5000
          name: http
          protocol: TCP
--------------------------------------------------------------------
achal@LAPTOP-# cat web-svc.yml
apiVersion: v1
kind: Service
metadata:
  name: web
  labels:
    name: web
    app: demo
spec:
  selector:
    name: web
  type: NodePort
  ports:
   - port: 80
     name: http
     targetPort: 5000
     protocol: TCP
--------------------------------------------------------------------
achal@LAPTOP-#:Deploy$ cat db-pod.yml
apiVersion: "v1"
kind: Pod
metadata:
  name: mysql
  labels:
    name: mysql
    app: demo
spec:
  containers:
    - name: mysql
      image: mysql:5.7.25
      ports:
```

```
        - containerPort: 3306
          protocol: TCP
      env:
        -
          name: "MYSQL_ROOT_PASSWORD"
          value: "password"
--------------------------------------------------------------------
achal@LAPTOP-#:Deploy$ cat db-svc.yml
apiVersion: v1
kind: Service
metadata:
  name: mysql
  labels:
    name: mysql
    app: demo
spec:
  ports:
  - port: 3306
    name: mysql
    targetPort: 3306
  selector:
    name: mysql
    app: demo
--------------------------------------------------------------------
achal@LAPTOP-# kubectl get pods
NAME                        READY   STATUS    RESTARTS   AGE
mysql                       1/1     Running   0          1om
web1                        2/2     Running   0          50s
--------------------------------------------------------------------
achal@LAPTOP-# kubectl get svc
NAME    TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
mysql   ClusterIP   10.101.187.146   <none>        3306/TCP       5m47s
web     NodePort    10.98.202.95     <none>        80:31992/TCP   7m59s
--------------------------------------------------------------------
achal@LAPTOP-# curl http://localhost:8080/init
DB Init doneachal@LAPTOP-HPUOTJF1:~$

achal@LAPTOP-# curl http://localhost:8080/init
DB Init doneachal@LAPTOP-curl -i -H "Content-Type: application/json" -X POST -
d '{"uid": "1",{"uid": "1",
"user":"John Doe"}' http://localhost:8080/users/add
HTTP/1.0 200 OK
Content-Type: application/json
Content-Length: 5
Server: Werkzeug/1.0.1 Python/2.7.15
Date: Thu, 03 Sep 2026 09:57:59 GMT
--------------------------------------------------------------------
Addedachal@LAPTOP-# curl -i -H "Content-Type: application/json" -X POST -d
'{"uid": "2", "2",
"user":"Jane Doe"}' http://localhost:8080/users/add
HTTP/1.0 200 OK
Content-Type: application/json
Content-Length: 5
Server: Werkzeug/1.0.1 Python/2.7.15
Date: Thu, 03 Sep 2026 09:58:46 GMT
--------------------------------------------------------------------
Addedachal@LAPTOP-# curl -i -H "Content-Type: application/json" -X POST -d
'{"uid": "3", "3",
"user":"Bill Colls"}' http://localhost:8080/users/add
HTTP/1.0 200 OK
```

```
Content-Type: application/json
Content-Length: 5
Server: Werkzeug/1.0.1 Python/2.7.15
Date: Thu, 03 Sep 2026 09:59:24 GMT
----------------------------------------------------------------------
Addedachal@LAPTOP-# curl -i -H "Content-Type: application/json" -X POST -d
'{"uid": "4", "4",
"user":"Mike Taylor"}' http://localhost:8080/users/add
HTTP/1.0 200 OK
Content-Type: application/json
Content-Length: 5
Server: Werkzeug/1.0.1 Python/2.7.15
Date: Thu, 03 Sep 2026 09:59:50 GMT
----------------------------------------------------------------------
Addedachal@LAPTOP-# curl http://localhost:8080/users/1ers/1
John Doeachal@LAPTOP-HPUOTJF1:~$

chal@LAPTOP-# kubectl exec -it mysql -n cloudethix-achal -- mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 5
Server version: 5.7.25 MySQL Community Server (GPL)

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> show databases:
    -> ;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual
that corresponds to your MySQL server version for the right syntax to use near
':' at line 1
mysql> SHOW DATABASES;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| USERDB             |
| mysql              |
| performance_schema |
| sys                |
+--------------------+
5 rows in set (0.01 sec)

mysql> use USERDB;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> show tables;
+------------------+
| Tables_in_USERDB |
+------------------+
| users            |
+------------------+
1 row in set (0.00 sec)
```

```
mysql> select * from users;
+------+-------------+
| ID   | USER        |
+------+-------------+
|    1 | John Doe    |
|    2 | Jane Doe    |
|    3 | Bill Colls  |
|    4 | Mike Taylor |
+------+-------------+
4 rows in set (0.00 sec)
```

Que 5 →

- Create 1 Public Docker Hub registry named cloudethix_Initcontainer_yourname.
- Clone below repository on your system.

https://github.com/janakiramm/simpleapp.git

- Initialize a local repository & copy the code from above repo to your local repository in any of your working branch.
- Once code is copied , go to the Build directory and build docker image from docker file and add meaningful tags and push to docker hub repository.
- Once Images are pushed to Docker hub registries, create a directory named kube.

Inside the kube directory create deployement.yaml file with 3 replication , label app: simpleapp-webapp , containerPort: 80 and add the image that you have pushed in Docker Hub registry.

- Create one service.yaml file with type nodeport & select simpleapp-webapp pod with port 80 & targetPort 80 with any nodePort between range 30000-32768.
- Open the webpage in the browser and notice the changes and capture the snap.
- Then delete the deployment that you have just created.
- Update the deployment.yaml file and add volumeMounts with mountPath /usr/share/nginx/html from emptyDir: {} volume.
- Once above changes are added, add initContainers block with below parameters. Also add volumeMounts for Init Container with mountPath "/work-dir" from emptyDir: {} volume.

```
initContainers:
- name: install
image: busybox:1.28
command:
- wget
- "-O"
- "/work-dir/index.html"
- http://info.cern.ch
volumeMounts:
- name: workdir
mountPath: "/work-dir"
```

- Add volumes with emptyDir: {} in deployment.yaml file.
- Once the deployment.yaml file is ready, create the deployment & access the page in the browser and notice the changes.
- Prepare a well formatted document and write your understanding step by step.

```
achal@LAPTOP-# git clone https://github.com/janakiramm/simpleapp.git
Cloning into 'simpleapp'...
```

```
remote: Enumerating objects: 47, done.
remote: Total 47 (delta 0), reused 0 (delta 0), pack-reused 47 (from 1)
Receiving objects: 100% (47/47), 8.20 KiB | 1.64 MiB/s, done.
Resolving deltas: 100% (9/9), done.
-----------------------------------------------------------------
achal@LAPTOP-#:local-repo$ ls
Dockerfile  html  wrapper.sh
achal@LAPTOP-HPUOTJF1:local-repo$ git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this
warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
Initialized empty Git repository in /home/achal/Assignment-practice-3/task-
5/local-repo/.git/
-----------------------------------------------------------------
achal@LAPTOP-#$ podman images
REPOSITORY                                          TAG               IMAGE
ID      CREATED        SIZE
docker.io/achalpadol/cloudethix_multicontainer_achal  v1
7687bb45809e  2 hours ago    712 MB
-----------------------------------------------------------------
achal@LAPTOP-# podman push
docker.io/achalpadol/cloudethix_initcontainer_achal:v1
Getting image source signatures
Copying blob 5310eb16bf43 skipped: already exists
Copying blob a44b5c8be516 skipped: already exists
Copying blob c13f294dea34 skipped: already exists
-----------------------------------------------------------------
achal@LAPTOP-# cat deployment.yml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: simpleapp-webapp-1
spec:
  replicas: 3
  selector:
    matchLabels:
      app: simpleapp-webapp-1
  template:
    metadata:
      labels:
        app: simpleapp-webapp-1
    spec:
      containers:
        - name: simpleapp-webapp-1
          image: docker.io/achalpadol/cloudethix_initcontainer_achal:v1
          ports:
            - containerPort: 80
```

```
------------------------------------------------------------------------
achal@LAPTOP-# k get deployment
NAME                 READY   UP-TO-DATE   AVAILABLE   AGE
simpleapp-webapp-1   3/3     3            3           3m35s
------------------------------------------------------------------------
achal@LAPTOP-#k get pod
NAME                                  READY   STATUS    RESTARTS   AGE
simpleapp-webapp-1-6c56c976fc-5mpfj   1/1     Running   0          3m52s
simpleapp-webapp-1-6c56c976fc-cw5mh   1/1     Running   0          3m52s
simpleapp-webapp-1-6c56c976fc-xrmcv   1/1     Running   0          3m52s
------------------------------------------------------------------------
achal@LAPTOP-# cat service.yml
apiVersion: v1
kind: Service
metadata:
  name: simpleapp-service-1
spec:
  type: NodePort
  selector:
    app: simpleapp-webapp-1
  ports:
    - port: 80
      nodePort: 30009
------------------------------------------------------------------------
achal@LAPTOP-# k get svc
NAME                 TYPE       CLUSTER-IP    EXTERNAL-IP   PORT(S)
AGE
simpleapp-service-1   NodePort   10.98.49.99   <none>        80:30009/TCP
89s
------------------------------------------------------------------------
achal@LAPTOP-# curl http://127.0.0.1:44855
<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8">
  <title>Sample Deployment</title>
  <style>
    body {
      color: #ffffff;
      background-color: blue;
      font-family: Arial, sans-serif;
      font-size: 14px;
    }

    h1 {
      font-size: 500%;
      font-weight: normal;
      margin-bottom: 0;
    }

    h2 {
      font-size: 200%;
      font-weight: normal;
      margin-bottom: 0;
    }
  </style>
</head>
<body>
  <div align="center">
    <h1>Welcome to V1 of the web application</h1>
```

```
    <h2>This application will be deployed on Kubernetes.</h2>
  </div>
</body>
</html>
-----------------------------------------------------------------------
achal@LAPTOP-# k delete deployment simpleapp-webapp-1
deployment.apps "simpleapp-webapp-1" deleted from cloudethix-achal namespace
achal@LAPTOP-HPUOTJF1:kube$ k get deployment
No resources found in cloudethix-achal namespace.
-----------------------------------------------------------------------
# Update deplpyment.yml file
achal@LAPTOP-# cat deployment.yml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: simpleapp-webapp
  namespace: cloudethix-achal

spec:
  replicas: 3

  selector:
    matchLabels:
      app: simpleapp-webapp-1

  template:
    metadata:
      labels:
        app: simpleapp-webapp-1

    spec:

      initContainers:
      - name: install
        image: busybox:1.28
        command:
        - wget
        - "-O"
        - "/work-dir/index.html"
        - http://info.cern.ch
        volumeMounts:
        - name: workdir
          mountPath: "/work-dir"

      containers:
      - name: simpleapp-webapp-1
        image: docker.io/achalpadol/cloudethix_initcontainer_achal:v1
        ports:
        - containerPort: 80

        volumeMounts:
        - name: workdir
          mountPath: /usr/share/nginx/html

      volumes:
      - name: workdir
        emptyDir: {}
-----------------------------------------------------------------------
achal@LAPTOP-# k apply -f deployment.yml
deployment.apps/simpleapp-webapp-1 created
```

```
achal@LAPTOP-# k get deployment
NAME                 READY   UP-TO-DATE   AVAILABLE   AGE
simpleapp-webapp-1   3/3     3            3           10s
achal@LAPTOP-# k get pod

NAME                                  READY   STATUS    RESTARTS   AGE
simpleapp-webapp-1-5dc7b6d554-4cjnh   1/1     Running   0          14s
simpleapp-webapp-1-5dc7b6d554-qlzzq   1/1     Running   0          14s
simpleapp-webapp-1-5dc7b6d554-xfkpx   1/1     Running   0          14s
-----------------------------------------------------------------------
achal@LAPTOP-# curl http://127.0.0.1:46571
<html><head></head><body><header>
<title>http://info.cern.ch</title>
</header>

<h1>http://info.cern.ch - home of the first website</h1>
<p>From here you can:</p>
<ul>
<li><a href="http://info.cern.ch/hypertext/WWW/TheProject.html">Browse the
first website</a></li>
<li><a href="http://line-
mode.cern.ch/www/hypertext/WWW/TheProject.html">Browse the first website using
the line-mode browser simulator</a></li>
<li><a href="http://home.web.cern.ch/topics/birth-web">Learn about the birth
of the web</a></li>
<li><a href="http://home.web.cern.ch/about">Learn about CERN, the physics
laboratory where the web was born</a></li>
</ul>
</body></html>
-----------------------------------------------------------------------
Que 6 →
● Create 1 Public Docker Hub registry named cloudethix_hpa_yourname.
● Clone below repository on your system.
https://github.com/vivekamin/kubernetes-hpa-example.git
● Initialize a local repository & copy the code from above repo to your local
repository in
any of your working branch.
● Once code is copied , build a docker image from the docker file and add
meaningful tags
and push to the docker hub repository.
● Once the image is pushed, go to k8s directory and update deployment.yaml
file with
image name from your repo. And then create it.
● Open service.yml and change the type to nodePort and apply the same.
● Open the HPA.yaml file, notice it and then apply the same.
● Open the browser, and access the webpage.
● Now it's time to test the HPA working with the below command.
# kubectl run -i --tty load-generator --rm --image=busybox --restart=Never --
/bin/sh -c "while sleep 0.01; do wget -q -O- http://NODE_PORT_SERVICE_NAME;
done"
● Check the HPA from kubectl command and also check if the deployment is
scaling up.
● Take the snap , prepare a well formatted doc and write your understanding.
-----------------------------------------------------------------------
achal@LAPTOP-# git clone https://github.com/vivekamin/kubernetes-hpa-
example.git
Cloning into 'kubernetes-hpa-example'...
remote: Enumerating objects: 26, done.
remote: Total 26 (delta 0), reused 0 (delta 0), pack-reused 26 (from 1)
```

```
Receiving objects: 100% (26/26), done.
Resolving deltas: 100% (9/9), done.
------------------------------------------------------------------------
achal@LAPTOP-# git init
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this
warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName
false"
Initialized empty Git repository in /home/achal/Assignment-practice-3/task-
6/local-repo/.git/
------------------------------------------------------------------------
achal@LAPTOP-# sudo podman build -t
docker.io/achalpadol/cloudethix_hpa_achal:v1 .
STEP 1/6: FROM node:8.12.0-alpine

achal@LAPTOP-HPUOTJF1:cloudethix_hpa_achal$ sudo podman images
REPOSITORY                                  TAG            IMAGE ID
CREATED         SIZE
docker.io/achalpadol/cloudethix_hpa_achal  v1             f83e042dbcd5  45
seconds ago  69.2 MB
------------------------------------------------------------------------
achal@LAPTOP-# cat deployment.yml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: node-example
spec:
  replicas: 1

  selector:
    matchLabels:
      app: node-example

  template:
    metadata:
      labels:
        app: node-example
    spec:
      containers:
      - name: node-example
        image: docker.io/achalpadol/cloudethix_hpa_achal:v1
        imagePullPolicy: Always
        ports:
        - containerPort: 3000
        resources:
            limits:
              cpu: "0.5"
            requests:
```

```
              cpu: "0.25"
--------------------------------------------------------------------------
achal@LAPTOP-# k apply -f deployment.yml
deployment.apps/node-example created
achal@LAPTOP-HPUOTJF1:local-repo$ k get pod
NAME                            READY   STATUS    RESTARTS   AGE
node-example-68f464997b-29p9h   0/1     Pending   0          8s
--------------------------------------------------------------------------
achal@LAPTOP-# cat service.yml
apiVersion: v1
kind: Service
metadata:
  name: node-example
  labels:
    app: node-example
spec:
  selector:
    app: node-example
  ports:
  - port: 3000
    protocol: TCP
    nodePort: 30001
  type: NodePort
--------------------------------------------------------------------------
achal@LAPTOP-# k apply -f service.yml
service/node-example created
achal@LAPTOP-HPUOTJF1:local-repo$ k get svc
NAME                   TYPE        CLUSTER-IP     EXTERNAL-IP   PORT(S)
AGE
kubernetes             ClusterIP   10.96.0.1      <none>        443/TCP
3d16h
node-example           NodePort    10.103.61.92   <none>        3000:30001/TCP
4s
--------------------------------------------------------------------------
achal@LAPTOP-# k apply -f hpa.yml
horizontalpodautoscaler.autoscaling/node-example configured
achal@LAPTOP-HPUOTJF1:k8s$ k get hpa
NAME           REFERENCE                  TARGETS      MINPODS   MAXPODS
REPLICAS   AGE
node-example   Deployment/node-example   cpu: 0%/1%   1         4         1
3d
--------------------------------------------------------------------------
achal@LAPTOP-# minikube service node-example   --url
http://127.0.0.1:45331
❗  Because you are using a Docker driver on linux, the terminal needs to be
open to run it.

achal@LAPTOP-# curl http://127.0.0.1:45331
Hello World!!
--------------------------------------------------------------------------
^Cachal@LAPTOP-HPUOTJF1:k8s$ k top pod
NAME                            CPU(cores)   MEMORY(bytes)
node-example-68f464997b-29p9h   1m           27Mi
--------------------------------------------------------------------------
Que 7 →
● Create 1 Public Docker Hub registry named cloudethix_cronjob_yourname.
● Initialize a local repository & copy below code (three files) to your local
repository in any
of your working branch.
● Once code is copied, build the docker image from Dockerfile , add meaningful
```

tags and
then push the docker image to Docker hub registry.
● Now update the pythoncronjob.yml file to change the image name that you have just
pushed to docker hub registry.
● Now create a cron job using pythoncronjob.yml file. Check with kubectl command if the
cron job is created.
● Check the Job name which is created by cronjob from command line or lens.
● Then check the pod logs which are created by the job and capture the output.
● Prepare well formatted documents and write your understanding.

```
achal@LAPTOP-# cat Dockerfile
FROM python:3.7-alpine
#add user group and ass user to that group
RUN addgroup -S appgroup && adduser -S appuser -G appgroup
#creates work dir
WORKDIR /app
#copy python script to the container folder app
COPY helloworld.py /app/helloworld.py
RUN chmod +x /app/helloworld.py
#user is appuser
USER appuser
ENTRYPOINT ["python", "/app/helloworld.py"]
```

```
achal@LAPTOP-# docker build -t cloudethix_cronjob_achal:v4.
STEP 1/7: FROM python:3.7-alpine
Resolved "python" as an alias
(/etc/containers/registries.conf.d/shortnames.conf)
Trying to pull docker.io/library/python:3.7-alpine...
Getting image source signatures
```

```
chal@LAPTOP-# podman tag cloudethix_cronjob_achal:v4
docker.io/achalpadol/cloudethix_cronjob_achal:v4
achal@LAPTOP-HPUOTJF1:cloudethix_cronjob_achal$ podman images
REPOSITORY                                          TAG               IMAGE
ID      CREATED        SIZE
localhost/cloudethix_cronjob_achal                  v4
0e0635316d06  2 minutes ago  49.5 MB
docker.io/achalpadol/cloudethix_cronjob_achal       v4
0e0635316d06  2 minutes ago  49.5 MB
```

```
achal@LAPTOP-# podman push docker.io/achalpadol/cloudethix_cronjob_achal:v4
Getting image source signatures
Copying blob c43d3c0ede05 done   |
Copying blob 148762f75a1f skipped: already exists
Copying blob 4819c95424fc skipped: already exists
Copying blob 9875af95546d skipped: already exists
Copying blob ea1518237b37 skipped: already exists
```

```
achal@LAPTOP-# k apply -f pythoncronjob.yml
cronjob.batch/python-helloworld created
achal@LAPTOP-HPUOTJF1:task-7$ k get pod
NAME                                READY   STATUS      RESTARTS   AGE
python-helloworld-29812643-8cfhn    0/1     Completed   0          2m11s
python-helloworld-29812644-grh59    0/1     Completed   0          71s
python-helloworld-29812645-xhzbj    0/1     Completed   0          11s
```
